Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Manuel & Maria Silva Dairy

Physical address of dairy:

Number and Street	City	County	Zip Code
15420 E Highway 4	Stockton	San Joaquin	95215

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/15/1945

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X183-X150-XX01-XXXX

B. OPERATORS

Silva, Manuel

Operator name: Silva, Manuel

Telephone no.: (209) 948-4561 Landline, (209) 564-7297 Cellular

Mailing Address Number and Street	City	State	Zip Code
21487 White RD	Turlock	CA	95380

Silva, Maria

Operator name: Silva, Maria

Telephone no.: (209) 948-4561 Landline, (209) 390-3259 Cellular

Mailing Address Number and Street	City	State	Zip Code
21487 White RD	Turlock	CA	95380

This operator is responsible for paying permit fees.

C. OWNERS

Togninali Trust

Legal owner name: Togninali Trust

Telephone no.: (209) 948-0494 Landline, (209) 401-8923 Cellular

Mailing Address Number and Street	City	State	Zip Code
14500 E Highway 4	Stockton	CA	95215

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	400	0	0	0
Number under roof	0	0	0	300	0	0
Maximum number	0	0	400	300	0	0
Average number	0	0	400	300	0	0
Avg live weight (lbs)	0	0	1,100	800		

Predominant milk cow breed: Jersey-Holstein Cross

Average milk production: 1 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 7,251.79 *tons per reporting period*

Total nitrogen from manure: 66,430.00 *lbs per reporting period*

After ammonia losses (30% loss applied): 46,501.00 *lbs per reporting period*

Total phosphorus from manure: 11,242.00 *lbs per reporting period*

Total potassium from manure: 1.00 *lbs per reporting period*

Total salt from manure: 0.00 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 5,400,000 *gallons*

Total nitrogen generated: 11,941.70 *lbs*

Total phosphorus generated: 1,668.68 *lbs*

Total potassium generated: 280,144.73 *lbs*

Total salt generated: 168,265.24 *lbs*

	5,400,000	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	5,400,000	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Dom. Well	Ground water
East Irr. Well	Ground water
West Irr. Well	Ground water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

Date	Material type / Description	Quantity	Reporting basis	Moisture (%)	N (%)	P (%)	K (%)	Salt (%)
07/01/2023	Solid commercial fertilizer *Commercial Fertilizer Side dress*	5.60 *ton*	Dry-weight	8.0	92.000000	0.000000	0.000000	0.000000

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Commercial fertilizer / Other	9,479.68	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Total imports for all materials	9,479.68	0.00	0.00	0.00

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

No liquid nutrient exports entered.

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
Field #M1	80	80	4	both	X183-X150-XX01-XXXX
Field #M2	55	55	4	both	X183-X150-XX01-XXXX
Field #M3	15	15	2	both	X183-X150-XX01-XXXX
Totals for areas that were used for application	150	150	10		
Totals for areas that were not used for application					
Land application area totals	150	150	10		

B. CROPS AND HARVESTS

Field #M1

Field name: Field #M1

10/21/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 80 Plant date: 10/21/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/26/2023	560.00 *ton*	Dry-weight		52.6	10,100.00	2,340.00	14,700.00		12.00
04/26/2023	560.00 *ton*	Dry-weight		51.8	8,790.00	2,360.00	14,900.00		11.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	153.60	35.20	220.80	0.00
Total actual harvest content	14.00	126.34	31.45	198.09	1,585.84

Field #M1

05/08/2023: Corn, silage

Crop: Corn, silage Acres planted: 80 Plant date: 05/08/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/08/2023	1,200.00 *ton*	Dry-weight		68.0	10,700.00	2,660.00	13,900.00		5.90
09/08/2023	1,200.00 *ton*	Dry-weight		71.0	12,000.00	2,300.00	15,000.00		5.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	47.60	246.40	0.00
Total actual harvest content	30.00	207.12	45.55	263.94	1,071.00

Field #M2

Field name: Field #M2

10/21/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 55 Plant date: 10/21/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/27/2023	560.00 *ton*	Dry-weight		46.8	10,900.00	2,240.00	14,800.00		12.80
04/27/2023	210.00 *ton*	Dry-weight		53.0	9,090.00	2,150.00	14,000.00		11.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	185.60	36.80	249.60	0.00
Total actual harvest content	14.00	150.71	31.98	210.58	1,813.78

Field #M2

05/08/2023: Corn, silage

Crop: Corn, silage Acres planted: 55 Plant date: 05/08/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/08/2023	1,200.00 *ton*	Dry-weight		68.4	10,500.00	2,750.00	14,000.00		6.10
09/08/2023	450.00 *ton*	Dry-weight		63.8	9,900.00	3,330.00	14,300.00		6.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	47.60	246.40	0.00
Total actual harvest content	30.00	203.43	57.65	277.76	1,220.25

Field #M3

Field name: Field #M3

10/21/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 15 Plant date: 10/21/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/28/2023	210.00 *ton*	Dry-weight		53.0	9,090.00	2,150.00	14,000.00		11.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	14.00	119.62	28.29	184.24	1,566.04

05/08/2023: Corn, silage

Crop: Corn, silage Acres planted: 15 Plant date: 05/08/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/08/2023	450.00 *ton*	Dry-weight		63.8	9,900.00	3,330.00	14,300.00		6.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	198.80	67.20	252.00	0.00
Total actual harvest content	30.00	215.03	72.33	310.60	1,390.08

NUTRIENT BUDGET

A. LAND APPLICATIONS

Field #M1 - 10/21/2022: Oats, silage-soft dough

Field name: Field #M1

Crop: Oats, silage-soft dough — Plant date: 10/21/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/18/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure Application	Corral solids	146.88	36.89	83.81	0.00	720.00 *ton*
Application event totals		146.88	36.89	83.81	0.00	

Field #M1 - 05/08/2023: Corn, silage

Field name: Field #M1

Crop: Corn, silage — Plant date: 05/08/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
West Irr. Well	Ground water	6.87	0.00	0.00	0.00	4,950,000.00 *gal*
East Irr. Well	Ground water	3.17	0.00	0.00	0.00	3,960,000.00 *gal*
Application event totals		10.04	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
West Irr. Well	Ground water	6.61	0.00	0.00	0.00	4,762,500.00 *gal*
East Irr. Well	Ground water	3.05	0.00	0.00	0.00	3,810,000.00 *gal*
Application event totals		9.66	0.00	0.00	0.00	

Field #M1 - 05/08/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
LAGOON APPLICATION	Process wastewater	28.74	0.00	0.21	485.43	840,000.00 *gal*
East Irr. Well	Ground water	3.05	0.00	0.00	0.00	3,810,000.00 *gal*
West Irr. Well	Ground water	6.61	0.00	0.00	0.00	4,762,000.00 *gal*
Application event totals		38.40	0.00	0.21	485.43	

07/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
East Irr. Well	Ground water	2.93	0.00	0.00	0.00	3,660,000.00 *gal*
West Irr. Well	Ground water	6.35	0.00	0.00	0.00	4,575,000.00 *gal*
Application event totals		9.28	0.00	0.00	0.00	

07/15/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Commercial Fertilizer Sidedress	Solid commercial fertilizer	75.00	0.00	0.00	0.00	
Application event totals		75.00	0.00	0.00	0.00	

07/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
West Irr. Well	Ground water	6.35	0.00	0.00	0.00	4,575,000.00 *gal*
East Irr. Well	Ground water	2.93	0.00	0.00	0.00	3,660,000.00 *gal*
Application event totals		9.28	0.00	0.00	0.00	

08/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	49.90	11.09	1,861.85	620.62	2,010,000.00 *gal*
West Irr. Well	Ground water	4.79	0.00	0.00	0.00	3,450,000.00 *gal*
East Irr. Well	Ground water	2.21	0.00	0.00	0.00	2,760,000.00 *gal*
Application event totals		56.90	11.09	1,861.85	620.62	

Field #M1 - 05/08/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
West Irr. Well	Ground water	6.35	0.00	0.00	0.00	4,575,000.00 *gal*
East Irr. Well	Ground water	2.93	0.00	0.00	0.00	3,660,000.00 *gal*
Application event totals		9.28	0.00	0.00	0.00	

Field #M2 - 10/21/2022: Oats, silage-soft dough

Field name: Field #M2

Crop: Oats, silage-soft dough Plant date: 10/21/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/18/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure Application	Corral solids	146.88	36.89	83.81	0.00	495.00 *ton*
Application event totals		146.88	36.89	83.81	0.00	

Field #M2 - 05/08/2023: Corn, silage

Field name: Field #M2

Crop: Corn, silage Plant date: 05/08/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
West Irr. Well	Ground water	7.11	0.00	0.00	0.00	3,525,000.00 *gal*
East Irr. Well	Ground water	3.29	0.00	0.00	0.00	2,820,000.00 *gal*
Application event totals		10.40	0.00	0.00	0.00	

Field #M2 - 05/08/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
East Irr. Well	Ground water	3.08	0.00	0.00	0.00	2,640,000.00 *gal*
West Irr. Well	Ground water	6.66	0.00	0.00	0.00	3,300,000.00 *gal*
Application event totals		9.74	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
LAGOON APPLICATION	Process wastewater	29.86	0.00	0.22	504.34	600,000.00 *gal*
West Irr. Well	Ground water	6.66	0.00	0.00	0.00	3,300,000.00 *gal*
East Irr. Well	Ground water	3.08	0.00	0.00	0.00	2,640,000.00 *gal*
Application event totals		39.60	0.00	0.22	504.34	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
East Irr. Well	Ground water	3.08	0.00	0.00	0.00	2,640,000.00 *gal*
West Irr. Well	Ground water	6.66	0.00	0.00	0.00	3,300,000.00 *gal*
Application event totals		9.74	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
West Irr. Well	Ground water	6.28	0.00	0.00	0.00	3,112,500.00 *gal*
East Irr. Well	Ground water	2.90	0.00	0.00	0.00	2,490,000.00 *gal*
Application event totals		9.18	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/18/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
COMMERCIAL FERTILIZER SIDEDRESS	Solid commercial fertilizer	75.00	0.00	0.00	0.00	
Application event totals		75.00	0.00	0.00	0.00	

Field #M2 - 05/08/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	49.83	11.08	1,859.33	619.78	1,380,000.00 *gal*
West Irr. Well	Ground water	4.84	0.00	0.00	0.00	2,400,000.00 *gal*
East Irr. Well	Ground water	2.24	0.00	0.00	0.00	1,920,000.00 *gal*
Application event totals		56.91	11.08	1,859.33	619.78	

08/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
West Irr. Well	Ground water	6.28	0.00	0.00	0.00	3,112,500.00 *gal*
East Irr. Well	Ground water	2.90	0.00	0.00	0.00	2,490,000.00 *gal*
Application event totals		9.18	0.00	0.00	0.00	

Field #M3 - 10/21/2022: Oats, silage-soft dough

Field name: Field #M3

Crop: Oats, silage-soft dough Plant date: 10/21/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/18/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure Application	Corral solids	146.88	36.89	83.81	0.00	135.00 *ton*
Application event totals		146.88	36.89	83.81	0.00	

Field #M3 - 05/08/2023: Corn, silage

Field name: Field #M3

Crop: Corn, silage Plant date: 05/08/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Field #M3 - 05/08/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
West Irr. Well	Ground water	10.54	0.00	0.00	0.00	1,425,000.00 *gal*
East Irr. Well	Ground water	4.87	0.00	0.00	0.00	1,140,000.00 *gal*
Application event totals		15.41	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
East Irr. Well	Ground water	4.10	0.00	0.00	0.00	960,000.00 *gal*
West Irr. Well	Ground water	8.88	0.00	0.00	0.00	1,200,000.00 *gal*
Application event totals		12.98	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
LAGOON APPLICATION	Process wastewater	32.85	0.00	0.25	554.78	180,000.00 *gal*
West Irr. Well	Ground water	8.88	0.00	0.00	0.00	1,200,000.00 *gal*
East Irr. Well	Ground water	4.10	0.00	0.00	0.00	960,000.00 *gal*
Application event totals		45.83	0.00	0.25	554.78	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
West Irr. Well	Ground water	6.66	0.00	0.00	0.00	900,000.00 *gal*
East Irr. Well	Ground water	3.08	0.00	0.00	0.00	720,000.00 *gal*
Application event totals		9.74	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/16/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
COMMERCIAL FERTILIZER SIDEDRESS	Solid commercial fertilizer	75.00	0.00	0.00	0.00	
Application event totals		75.00	0.00	0.00	0.00	

Field #M3 - 05/08/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
East Irr. Well	Ground water	3.08	0.00	0.00	0.00	720,000.00 *gal*
West Irr. Well	Ground water	6.66	0.00	0.00	0.00	900,000.00 *gal*
Application event totals		9.74	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	51.64	11.48	1,926.69	642.23	390,000.00 *gal*
West Irr. Well	Ground water	5.55	0.00	0.00	0.00	750,000.00 *gal*
East Irr. Well	Ground water	2.56	0.00	0.00	0.00	600,000.00 *gal*
Application event totals		59.75	11.48	1,926.69	642.23	

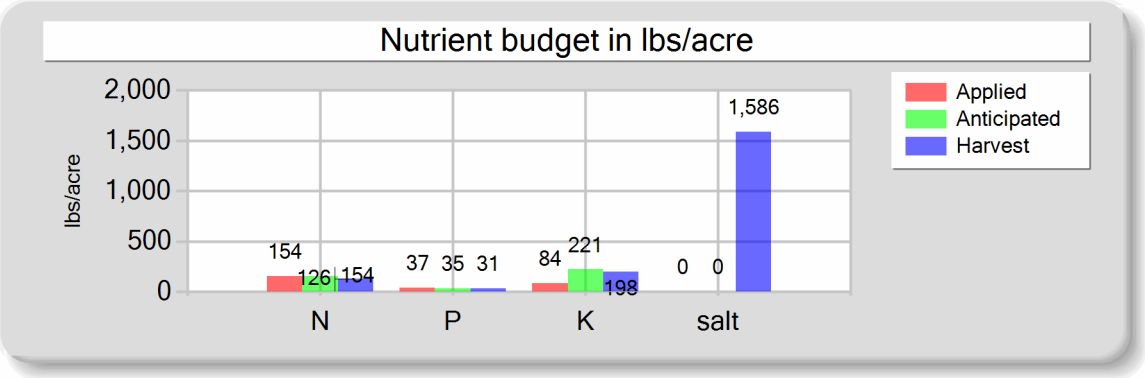
Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
West Irr. Well	Ground water	6.66	0.00	0.00	0.00	900,000.00 *gal*
East Irr. Well	Ground water	3.08	0.00	0.00	0.00	720,000.00 *gal*
Application event totals		9.74	0.00	0.00	0.00	

B. NUTRIENT BUDGET

Field #M1 - 10/21/2022: Oats, silage-soft dough

Field name: Field #M1 Crop: Oats, silage-soft dough Plant date: 10/21/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	146.88	36.89	83.81	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	153.88	36.89	83.81	0.00
Anticipated crop nutrient removal	153.60	35.20	220.80	0.00
Actual crop nutrient removal	126.34	31.45	198.09	1,585.84
Nutrient balance	27.54	5.44	-114.29	-1,585.84
Applied to removed ratio	1.22	1.17	0.42	0.00

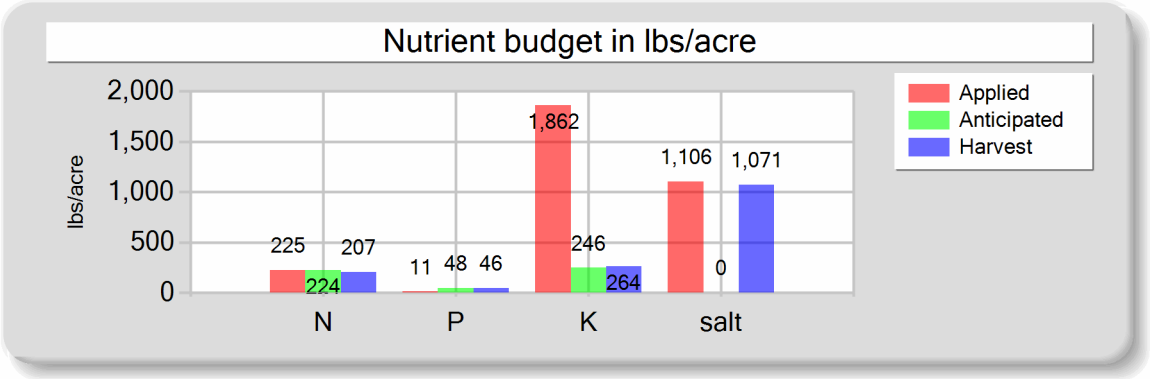
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
2 *harvests*

Field #M1 - 05/08/2023: Corn, silage

Field name: Field #M1 Crop: Corn, silage Plant date: 05/08/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	75.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	78.64	11.09	1,862.07	1,106.05
Fresh water	64.19	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	224.83	11.09	1,862.07	1,106.05
Anticipated crop nutrient removal	224.00	47.60	246.40	0.00
Actual crop nutrient removal	207.12	45.55	263.94	1,071.00
Nutrient balance	17.71	-34.45	1,598.13	35.05
Applied to removed ratio	1.09	0.24	7.05	1.03

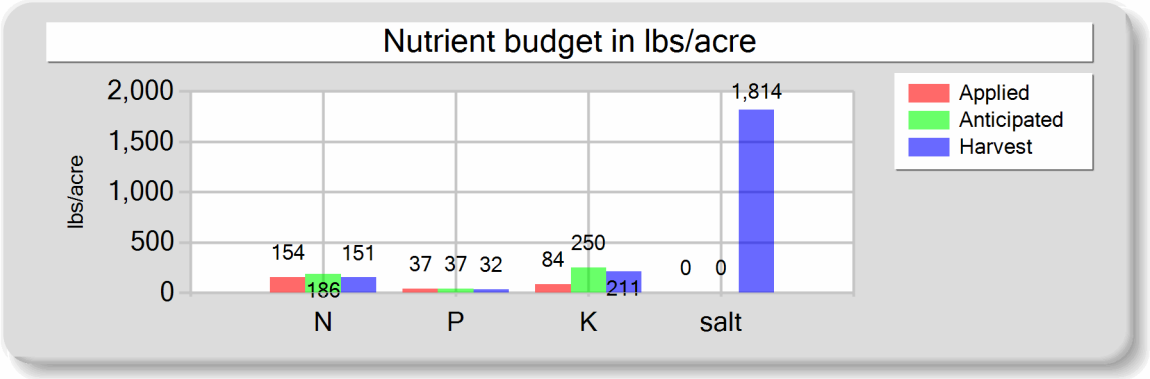
Fresh water applied	
56,969,500.00	*gallons*
2,097.99	*acre-inches*
26.22	*inches/acre*

Process wastewater applied	
2,850,000.00	*gallons*
104.96	*acre-inches*
1.31	*inches/acre*

Total harvests for the crop	
2	*harvests*

Field #M2 - 10/21/2022: Oats, silage-soft dough

Field name: Field #M2 Crop: Oats, silage-soft dough Plant date: 10/21/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	146.88	36.89	83.81	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	153.88	36.89	83.81	0.00
Anticipated crop nutrient removal	185.60	36.80	249.60	0.00
Actual crop nutrient removal	150.71	31.98	210.58	1,813.78
Nutrient balance	3.17	4.91	-126.77	-1,813.78
Applied to removed ratio	1.02	1.15	0.40	0.00

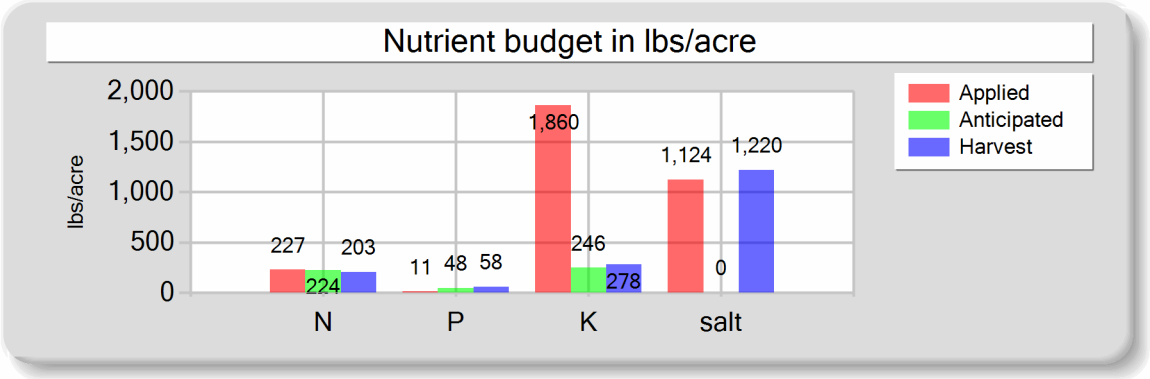
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
2 *harvests*

Field #M2 - 05/08/2023: Corn, silage

Field name: Field #M2 Crop: Corn, silage Plant date: 05/08/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	75.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	79.69	11.08	1,859.55	1,124.12
Fresh water	65.05	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	226.74	11.08	1,859.55	1,124.12
Anticipated crop nutrient removal	224.00	47.60	246.40	0.00
Actual crop nutrient removal	203.43	57.65	277.76	1,220.25
Nutrient balance	23.32	-46.57	1,581.79	-96.13
Applied to removed ratio	1.11	0.19	6.69	0.92

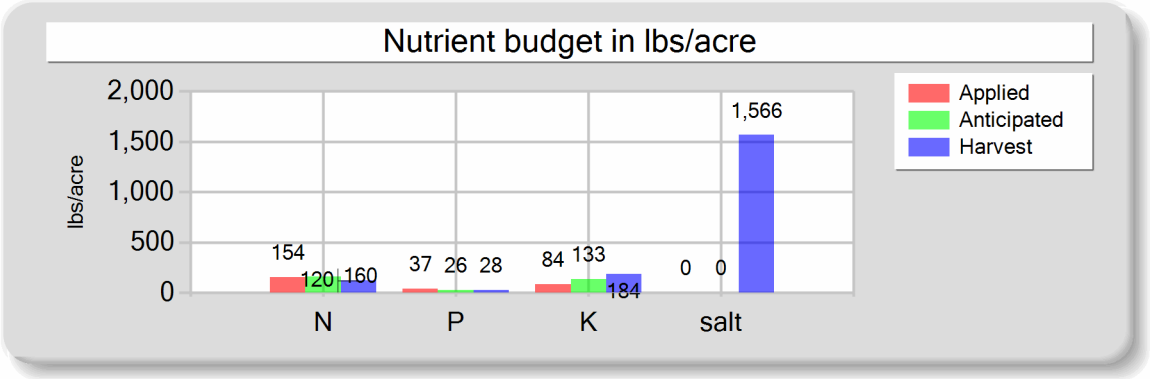
Fresh water applied	
39,690,000.00	*gallons*
1,461.65	*acre-inches*
26.58	*inches/acre*

Process wastewater applied	
1,980,000.00	*gallons*
72.92	*acre-inches*
1.33	*inches/acre*

Total harvests for the crop	
2	*harvests*

Field #M3 - 10/21/2022: Oats, silage-soft dough

Field name: Field #M3 Crop: Oats, silage-soft dough Plant date: 10/21/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	146.88	36.89	83.81	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	153.88	36.89	83.81	0.00
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	119.62	28.29	184.24	1,566.04
Nutrient balance	34.26	8.60	-100.43	-1,566.04
Applied to removed ratio	1.29	1.30	0.45	0.00

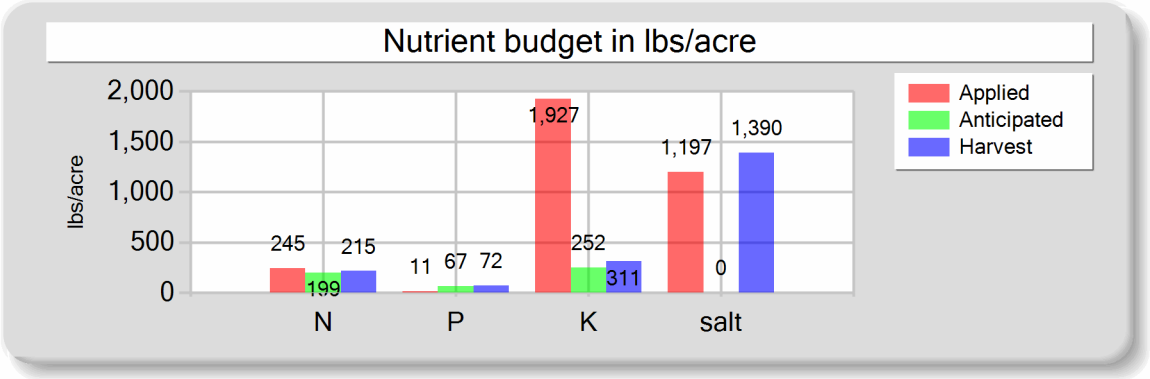
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

Field #M3 - 05/08/2023: Corn, silage

Field name: Field #M3 Crop: Corn, silage Plant date: 05/08/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	75.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	84.48	11.48	1,926.94	1,197.01
Fresh water	78.70	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	245.18	11.48	1,926.94	1,197.01
Anticipated crop nutrient removal	198.80	67.20	252.00	0.00
Actual crop nutrient removal	215.03	72.33	310.60	1,390.08
Nutrient balance	30.15	-60.85	1,616.34	-193.07
Applied to removed ratio	1.14	0.16	6.20	0.86

Fresh water applied	
13,095,000.00	*gallons*
482.24	*acre-inches*
32.15	*inches/acre*

Process wastewater applied	
570,000.00	*gallons*
20.99	*acre-inches*
1.40	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

T2168072A

Sample and source description: T2168072A

Sample date: 06/17/2022 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 52.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	17,000.00	4,270.00	9,700.00							0.00
DL	1.00	1.00	1.00							1.00

T3139082F

Sample and source description: T3139082F

Sample date: 05/19/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 60.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,300.00	3,240.00	5,790.00							0.00
DL	1.00	1.00	1.00							1.00

T3258036B

Sample and source description: T3258036B

Sample date: 09/15/2023 Material type: Scraped material Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 38.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	9,700.00	2,860.00	6,320.00	8,710.00	2,660.00	2,300.00	2,140.00	629.00		63.00
DL	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00		1.00

B. PROCESS WASTEWATER ANALYSES

T3053015A

Sample and source description: T3053015A

Sample date: 02/22/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	291.00	165.00	0.00	0.00	88.60	972.00								6,350.00	4,760
DL	1.00	1.00	1.00	1.00	1.00	1.00								100.00	1

T3139082E

Sample and source description: T3139082E

Sample date: 05/19/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	328.00	213.00	0.00	0.00	0.00	2.45								8,000.00	5,540
DL	1.00	1.00	1.00	1.00	2.00	1.00								100.00	1

T3258036A

Sample and source description: T3258036A

Sample date: 09/15/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.70

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	238.00	182.00	0.00	0.00	52.90	8,880.00	99.00	62.00	363.00	2,020.00	0.00	432.00	37.30	7,110.00	2,960
DL	1.00	1.00	1.00	0.14	1.00	1.00	1.00	1.00	1.00	1.00	2.00	1.00	1.00	100.00	1

T3342029A

Sample and source description: T3342029A

Sample date: 12/28/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	540.00	328.00	0.00	0.00	50.40	844.00								6,660.00	8,220
DL	1.00	1.00	1.00	1.00	1.00	1.00								100.00	1

C. FRESH WATER ANALYSES

Dom. Well

T3258036E

Sample description: T3258036E

Sample date: 09/15/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	7.82	0.00	7.82								467.00	0
DL	1.00	0.50	1.00								100.00	1

East Irr. Well

T3258036D

Sample description: T3258036D

Sample date: 09/15/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	7.68	0.00	7.68								421.00	0
DL	1.00	0.50	1.00								100.00	1

West Irr. Well

T3258036C

Sample description: T3258036C

Sample date: 09/15/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	13.30	0.00	13.30								632.00	0
DL	1.00	0.50	1.00								1.00	1

D. SOIL ANALYSES

No soil analyses entered.

E. PLANT TISSUE ANALYSES

Field #M1 - 10/21/2022: Oats, silage-soft dough

T3139082A

Sample and source description: T3139082A

Sample date: 05/19/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 52.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	10,100.00	2,340.00	14,700.00		12.00
DL	1.00	1.00	1.00		1.00

T3139082B

Sample and source description: T3139082B

Sample date: 05/19/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 51.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	8,790.00	2,360.00	14,900.00		11.70
DL	1.00	1.00	1.00		1.00

Field #M1 - 05/08/2023: Corn, silage

T3258036F

Sample and source description: T3258036F

Sample date: 09/15/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 68.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	10,700.00	2,660.00	13,900.00		5.90
DL	1.00	1.00	1.00		1.00

Field #M1 - 05/08/2023: Corn, silage

T3258036G

Sample and source description: T3258036G

Sample date: 09/15/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 71.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,000.00	2,300.00	15,000.00		5.80
DL	1.00	1.00	1.00		1.00

Field #M2 - 10/21/2022: Oats, silage-soft dough

T3139082C

Sample and source description: T3139082C

Sample date: 05/19/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 46.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	10,900.00	2,240.00	14,800.00		12.80
DL	1.00	1.00	1.00		1.00

T3139082D

Sample and source description: T3139082D

Sample date: 05/19/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 53.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	9,090.00	2,150.00	14,000.00		11.90
DL	1.00	1.00	1.00		1.00

Field #M2 - 05/08/2023: Corn, silage

Field #M2 - 05/08/2023: Corn, silage

T3258036H

Sample and source description: T3258036H

Sample date: 09/15/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 68.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	10,500.00	2,750.00	14,000.00		6.10
DL	1.00	1.00	1.00		1.00

T3258036I

Sample and source description: T3258036I

Sample date: 09/15/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 63.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	9,900.00	3,330.00	14,300.00		6.40
DL	1.00	1.00	1.00		1.00

Field #M3 - 10/21/2022: Oats, silage-soft dough

T3139082D

Sample and source description: T3139082D

Sample date: 05/19/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 53.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	9,090.00	2,150.00	14,000.00		11.90
DL	1.00	1.00	1.00		1.00

Field #M3 - 05/08/2023: Corn, silage

Field #M3 - 05/08/2023: Corn, silage

T3258036I

Sample and source description: T3258036I

Sample date: 09/15/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 63.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	9,900.00	3,330.00	14,300.00		6.40
DL	1.00	1.00	1.00		1.00

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

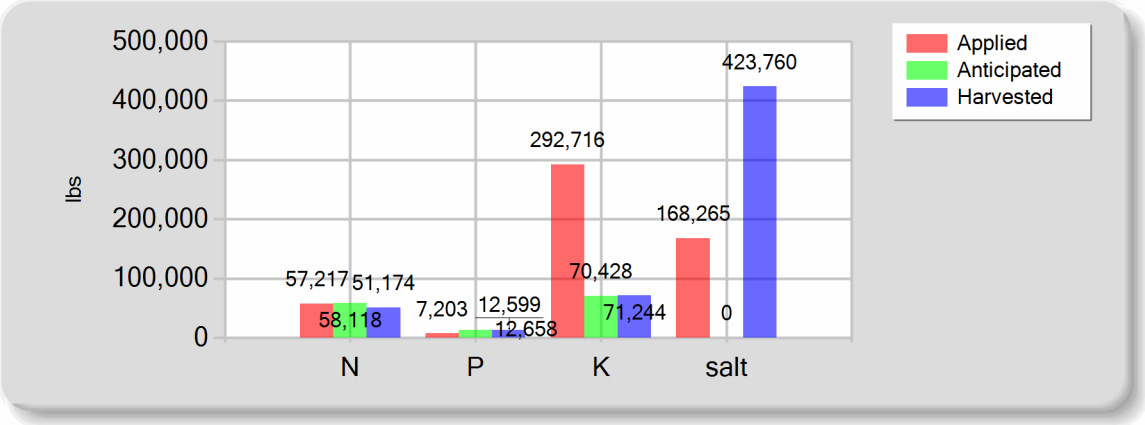
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

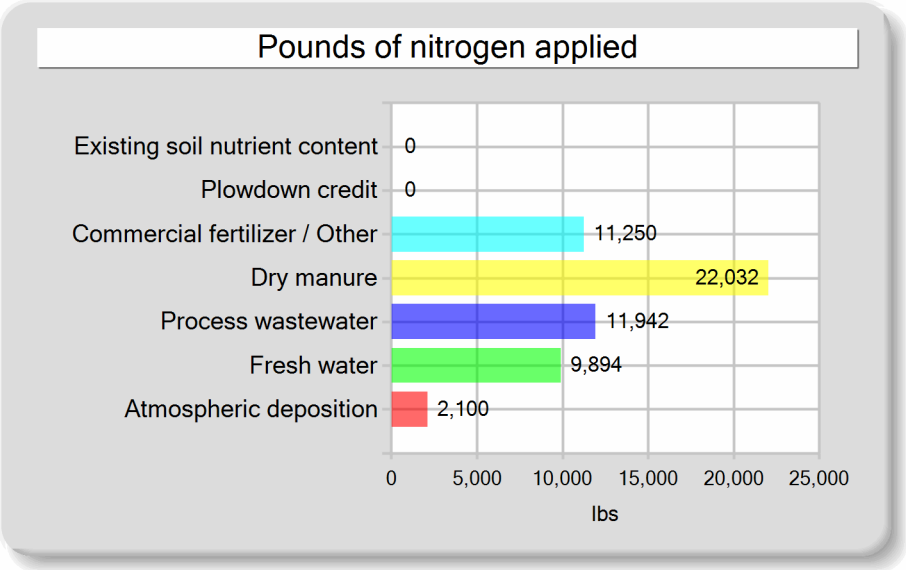
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

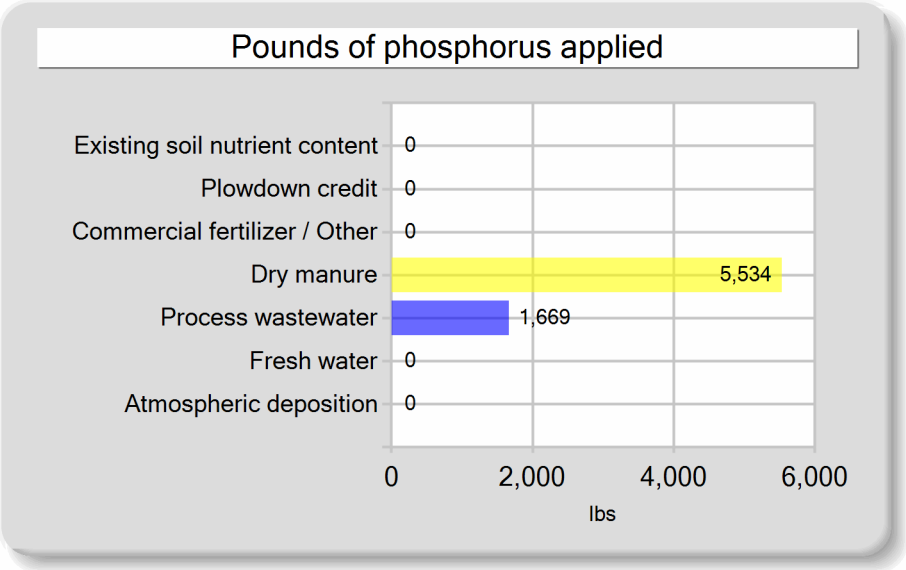
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	11,250.00	0.00	0.00	0.00
Dry manure	22,032.00	5,533.92	12,571.20	0.00
Process wastewater	11,941.70	1,668.68	280,144.73	168,265.24
Fresh water	9,893.76	0.00	0.00	0.00
Atmospheric deposition	2,100.00	0.00	0.00	0.00
Total nutrients applied	57,217.45	7,202.60	292,715.93	168,265.24
Anticipated crop nutrient removal	58,118.00	12,658.00	70,428.00	0.00
Actual crop nutrient removal	51,174.11	12,598.89	71,243.86	423,760.40
Nutrient balance	6,043.34	-5,396.29	221,472.07	-255,495.16
Applied to removed ratio	1.12	0.57	4.11	0.40

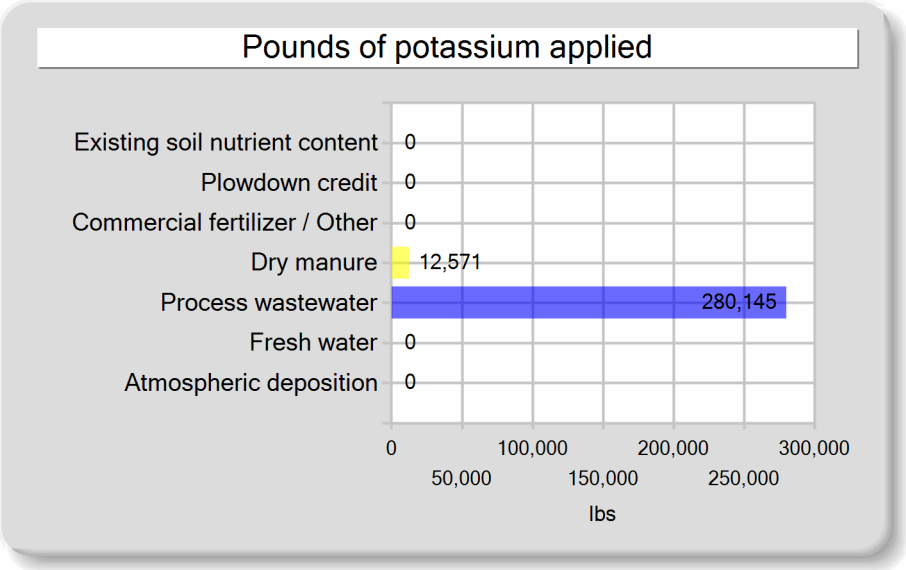
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

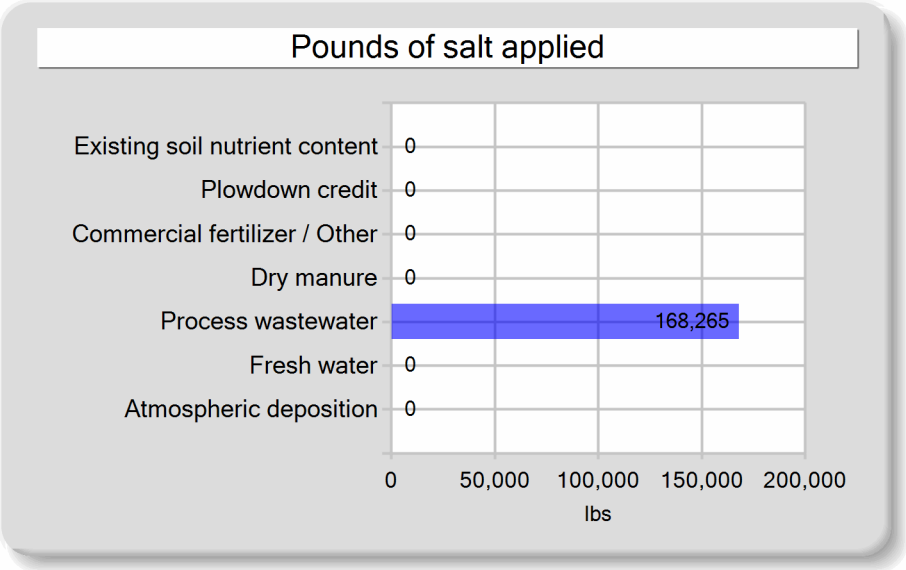


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	Yes
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

No notes entered for this annual report.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inqu of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are signific penalties for submitting false information, including the possibility of fine and imprisonment.

Gary Togninali	Manuel Silva
SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Togninali Trust	Manuel Silva
PRINT OR TYPE NAME	PRINT OR TYPE NAME
6-27-24	6-21-24
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

DENELE ANALYTICAL, INC.

865 S. KILROY
TURLOCK, CA 95380
PH# 209-634-9055 FAX# 209-634-9057

DAIRY COMPLIANCE

Client: Manuel & Maria Silva Dairy
Mailing address: 21487 White Rd.
(not facility address) Turlock, CA 95380
Phone#: (209)564-7297

Authorized Copy Released To:
JCRK, INC.
(209)327-0992
Sampled By: Maria Silva
Project ID:

ANALYSIS TO BE COMPLETED:

IRRIGATION/GROUND WATER
H1 EC*, NO3-N, NH4-N*
H6 EC, TN, TDS
H7 Combination of H1 & H6
H14 EC, NO3-N, NH4-N, Ca, Mg, Na, Bicarb Carb, SO4, Cl, TDS

PROCESS WASTE WATER
H3 EC*, NH4-N, TKN, TP, K, TDS
H4 EC*, NO3-N, NH4-N, TKN, TP, K, TDS
H5 Ca, Mg, Na, Bicarb, Carb, SO4, Cl
H8 Combination of H3 &H5
H9 Combination of H4 &H5

TILE DRAIN
H2 EC*, NO3-N, TP, NH4-N*, TDS

** Required Field Measurement*
___ (Check here if performed)

DISCHARGE WATER
E1 EC*, Temp*, pH*, NO3-N, NH4-N, TKN, TP, K, TDS, Dissolved Oxygen*, BOD 5, TSS, Total & Fecal Coliforms
E2 EC,* Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Dissolved Oxygen*, Total & Fecal Coliforms
E3 EC*, Temp*, pH*, NO3-N, NH4-N*, TP, Total & Fecal Coliforms
E4 EC*, Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Total & Fecal Coliform

MANURE
M1 % Moisture
M3 TN TP, K, % Moisture
M4 Ca, Mg, Na, Bicarb, Carb, SO4, Cl, Ash
M7 Combination of M3 & M4

PLANT TISSUE
F18 % Moisture, Ash, TN, P, K, DM and as received
F18+ % Moisture, Ash, TN, P, K, DM and as received (Package also includes NIR DCAD)
F19+ % Moisture, TN (Package includes complete plant tissue analysis)

SOIL
K3 Soluble Phosphorus
K4 1ft NO3, Organic Matter
K5 Pre-Plant (2ft NO3), Fall (1-2ft NO3), Fall (2-3ft NO3)
K6 0-1ft EC, NO3, Soluble P, K, Organic Matter
K7 Combination of K3 & K4
K8 Combination of K3 & K5

	Date Collected	Time Collected	Sample ID	Description	Temp	Well Purge	H1	H3	M3	F18	OTHER
1	9/13/23	11:30a	3rd Qtr Lagoon	Process wastewater		260'					H9+pH M7
2	✓	11:40a	Scraped Manure	Manure							
3	✓	12:10p	West Irr. Well	Ag Well	18.6°	20 min	X				
4	✓	12:30p	East Irr. Well	✓	18.0°	✓	X				
5	✓	1:00p	Dom. Well	Dom. well	18.2°	✓	X				
6	✓	1:15p	Field M-1 North	Corn Silage						X	
7	✓	1:30p	Field M-1 South	✓						X	
8	✓	1:45p	Field M-2	✓						X	
9	✓	2:00p	Fields M-2 + M-3	✓						X	
10											

Analysis

CHAIN OF CUSTODY RECORDING

	Signature	Company	Received Date & Time	Relinquished Date & Time
1st	Jacquie Captain	JCRK		9/15/23 8:00a
2nd		DENELE ANALYTICAL, INC.	9/15/23 1:30	

(LABORATORY USE ONLY)
Logged In By:
LABORATORY NUMBER: T3258036

Required to be submitted to any agency? (YES / NO)

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

www.denelelabs.com

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

Compliance Analysis Report

Customer
Manuel & Maria Silva Dairy
21487 White Rd.
Turlock, CA 95380

Grower

Consultant:
Jacquie Captein
P.O. Box 386
Herald, CA 95638

PURCHASE ORDER:
RECEIVED DATE: 9/15/23 1:30 PM
SUBMITTED BY: Jacquie
ANALYZED DATE: 9/18/23

Irrigation/ Ground Water (H1)
SOURCE: Irrigation Water

NOTES: Temperature was above qualifying at 18.6c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
West Irr. Well	T3258036C	Electrical Conductivity (EC)	EPA 120.1 and	0.632	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	13.3	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	405	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	98.2	94.0 (T3258022A)	94.6 (T3258022A)	0.635
Electrical Conductivity (EC)	97.6			1.87
Nitrate Nitrogen (NO3-N)	105	104 (T3258022A)	103 (T3258022A)	0.790

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

ELAP Certificate No.: 2714

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

www.denelelabs.com

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

Compliance Analysis Report

Customer
Manuel & Maria Silva Dairy
21487 White Rd.
Turlock, CA 95380

Grower

Consultant:
Jacquie Captein
P.O. Box 386
Herald, CA 95638

PURCHASE ORDER:
RECEIVED DATE: 9/15/23 1:30 PM
SUBMITTED BY: Jacquie
ANALYZED DATE: 9/18/23

Irrigation/ Ground Water (H1)
SOURCE: Irrigation Water

NOTES: Temperature was above qualifying at 18.0c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
East Irr. Well	T3258036D	Electrical Conductivity (EC)	EPA 120.1 and	0.421	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	7.68	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	270	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	98.2	94.0 (T3258022A)	94.6 (T3258022A)	0.635
Electrical Conductivity (EC)	97.6			1.87
Nitrate Nitrogen (NO3-N)	105	104 (T3258022A)	103 (T3258022A)	0.790

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

www.denelelabs.com

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

Compliance Analysis Report

Customer
Manuel & Maria Silva Dairy
21487 White Rd.
Turlock, CA 95380

Grower

Consultant:
Jacquie Captein
P.O. Box 386
Herald, CA 95638

PURCHASE ORDER:
RECEIVED DATE: 9/15/23 1:30 PM
SUBMITTED BY: Jacquie
ANALYZED DATE: 9/18/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 18.2c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Dom. Well	T3258036E	Electrical Conductivity (EC)	EPA 120.1 and	0.467	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	7.82	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	299	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	98.2	94.0 (T3258022A)	94.6 (T3258022A)	0.635
Electrical Conductivity (EC)	97.6			1.87
Nitrate Nitrogen (NO3-N)	105	104 (T3258022A)	103 (T3258022A)	0.790

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

ELAP Certificate No.: 2714